Worksheet 4A: Derivatives and Integrals — Inverse Operations — Solutions

AIML Foundations Mathematics
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Note to instructor: Answers below reflect the expected responses. Where a question asks for reasoning or explanation, sample responses are given. Accept any mathematically equivalent form unless the question specifies a particular form.

Part A

1. $f'(x) = 4x^3$; $\int x^4 dx = \frac{x^5}{5} + C$
2. $g'(t) = 5t^4$; $\int t^5 dt = \frac{t^6}{6} + C$
3. $h'(p) = 3p^2$; $\int p^3 dp = \frac{p^4}{4} + C$
4. $7x^6$; $\frac{x^8}{8} + C$
5. $2m$; $\frac{m^3}{3} + C$
6. $10r^9$; $\frac{r^{11}}{11} + C$
7. 1; $\frac{k^2}{2} + C$
8. $100x^{99}$; $\frac{x^{101}}{101} + C$
9. $-2x^{-3}$; $\frac{x^{-1}}{-1} + C = -\frac{1}{x} + C$
10. $-3t^{-4}$; $\frac{t^{-2}}{-2} + C = -\frac{1}{2t^2} + C$
11. $\frac{1}{2}x^{-1/2}$; $\frac{x^{3/2}}{3/2} + C = \frac{2}{3}x^{3/2} + C$
12. $\frac{1}{3}x^{-2/3}$; $\frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C$
13. $-\frac{1}{2}p^{-3/2}$; $\frac{p^{1/2}}{1/2} + C = 2\sqrt{p} + C$
14. $\frac{3}{2}n^{1/2}$; $\frac{n^{5/2}}{5/2} + C = \frac{2}{5}n^{5/2} + C$
15. Division by zero ($n + 1 = 0$)
16. x^3
17. $x^5 + C$
18. $6x^5$; $x^6 + C$; Yes, plus C
19. $\frac{x^5}{5} + C$; x^4; Yes exactly

Part B

20. All equal $2x$
21. Constants disappear when differentiating
22. same
23. x^3, x^3+1, x^3-100 (any three)
24. t^5, t^5+7, $t^5-\pi$ (any three)
25. p^4+C
26. x^6+C
27. $F(x)=x^3+5$
28. $G(t)=t^4+6$
29. $H(p)=p^2+p-6$
30. (a) $20t-5t^2+C$ (b) $C=5$ (c) $s(t)=20t-5t^2+5$
31. (a) $v(t)=6t$ (b) $s(t)=3t^2+10$

Part C

32. $15x^2$; $\frac{5x^4}{4}+C$
33. $-8t^3$; $-\frac{2t^5}{5}+C$
34. $3p^5$; $\frac{p^7}{14}+C$
35. 0; $7x+C$
36. 0; $-3x+C$
37. $3x^2+2x$; $\frac{x^4}{4}+\frac{x^3}{3}+C$
38. $4t^3-2t$; $\frac{t^5}{5}-\frac{t^3}{3}+C$
39. $6r+2$; r^3+r^2-5r+C
40. $12x^2-12x+2$; $x^4-2x^3+x^2-x+C$
41. $5k^4-9k^2+2$; $\frac{k^6}{6}-\frac{3k^4}{4}+k^2+C$
42. $8n^3+3n^2-10n+4$; $\frac{2n^5}{5}+\frac{n^4}{4}-\frac{5n^3}{3}+2n^2-7n+C$
43. $-3x^{-1}+C=-\frac{3}{x}+C$
44. $10\sqrt{t}+C$
45. $\frac{x^2}{2}-\frac{1}{x}+C$

Part D

46. $10x^4 - 12x^3 + 3x^2 - 7$
47. $t^3 - t^2 + t$
48. $h(p) = p^2 - p - 6$, $h'(p) = 2p - 1$
49. $y = x^2 - 2x + 1$, $y' = 2x - 2$
50. $x^3 - 2x^2 + 5x + C$
51. $\frac{t^5}{5} - \frac{t^4}{2} + \frac{t^2}{2} - 3t + C$
52. $m^6 - m^4 + m^2 + C$
53. $(u^2 + 1)^2 = u^4 + 2u^2 + 1$; $\frac{u^5}{5} + \frac{2u^3}{3} + u + C$
54. $4x^3 - 6x$
55. $r^4 - 3r^2 + C$
56. $6t^5 + 8t^3 - 1$
57. $k^5 - k^3 + k + C$
58. (a) $5x^4$ (b) $x^5 + C$ (c) Original plus constant
59. (a) $\frac{t^4}{4} - 2t^2 + C$ (b) $t^3 - 4t$ (c) Exactly $g(t)$
60. $f(x)$; $f(x) + C$

Part E

61. position
62. velocity
63. (a) $6t$ (b) $t^3 + C$ (c) t^3 (d) 8 m
64. (a) $3t^2 - 12t + 9$ (b) $6t - 12$ (c) $t = 1, 3$ (d) $t = 2$
65. $C(q) = 3q^2 + 10q + 500$
66. (b) $P(t) = 1000t + 25t^2 + 50000$ (c) $62,500$
67. (b) $W(t) = 100t - t^2$ (c) 900 L (d) $t = 50$ min

Part F

68. $\frac{8}{3}$
69. 8
70. 48
71. 0 (odd function, symmetric interval)
72. $\frac{14}{3}$

73. 12

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